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1 D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-
  PIKAN_Official\experiments\Volterra1D_MCPi.py
2 The network model is cPIKAN
3 Layers: ModuleList(
4   (0): ChebyKANLayer(in_features=1, out_features=10, degree=5)
5   (1): ChebyKANLayer(in_features=10, out_features=10, degree=5)
6   (2): ChebyKANLayer(in_features=10, out_features=1, degree=5)
7 )
8 Total trainable parameters: 720
9 Model on device: cuda:0
10 Start training...
11 Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
12 Training: 100%|██████████| 2000/2000 [00:09<00:00, 220.68epoch/s, Loss=1.
  29e-06, LR=1.0e-03, Best=1.29e-06]
13 Training finished. Best loss: 1.11e-06. Model saved to results/best_model.pth
14 Loss history saved to results/loss_history.npy
15 Training time: 9.855408191680908s
16 Relative loss: 0.001569639891386032
17 L2 loss: 0.0009987640660256147
18 Parameter containing:
19 tensor([[[ 0.0015, -0.0211,  0.3154,  0.0671,  0.1327, -0.2262],
20           [-0.1233, -0.0247,  0.0264,  0.2096,  0.1220, -0.0433],
21           [-0.0352,  0.0738,  0.0403,  0.3578,  0.0906, -0.1600],
22           [-0.2908, -0.0860, -0.4678,  0.0278,  0.1993,  0.0595],
23           [-0.0180,  0.1188,  0.1252, -0.1438,  0.1200,  0.2226],
24           [-0.0218,  0.2284,  0.2940,  0.4029,  0.2123, -0.1680],
25           [-0.0025,  0.0397,  0.2492, -0.0668, -0.0491,  0.1224],
26           [-0.4666,  0.0635,  0.0508,  0.1692, -0.1068,  0.1438],
27           [ 0.0345, -0.0674,  0.0264, -0.1679, -0.0501, -0.0568],
28           [-0.0981,  0.0701, -0.0297,  0.0225,  0.1587, -0.0186]]],
29         device='cuda:0', requires_grad=True)
30 Parameter containing:
31 tensor([[[ 3.5835e-02,  4.3597e-02, -3.2066e-02, -2.8146e-02,  1.4577e-02,
32           3.9802e-02],
33           [-1.4049e-03, -1.3495e-02,  1.3313e-02,  8.8128e-03, -2.5578e-03,
34           -4.1426e-02],
35           [ 4.1543e-02,  2.0691e-02, -9.4714e-03, -7.3274e-02, -1.7464e-03,
36           5.3575e-02],
37           [-1.0738e-02,  2.5562e-02, -1.6177e-02, -1.7965e-02, -9.3052e-03,
38           4.4808e-02],
39           [-1.1526e-02,  3.0299e-02,  2.7470e-02, -1.5887e-02,  1.9832e-02,
40           4.3969e-02],
41           [-7.1993e-03, -3.4418e-02,  3.5335e-02, -1.0537e-03, -1.3415e-02,
42           -2.8822e-02],
43           [-1.9653e-02, -4.3714e-03,  7.9978e-03, -4.5955e-02, -4.0380e-02,
44           -5.6790e-03],
45           [-3.0913e-02, -3.8797e-02, -3.6207e-03,  5.1079e-02,  4.1235e-02,
46           -9.2516e-02],
47           [-9.5518e-03,  1.5013e-02, -2.9706e-02, -1.1783e-02,  2.7821e-02,
48           3.5061e-02],
49           [-4.3198e-03, -8.4740e-02,  9.7127e-03,  4.2258e-02, -2.3149e-03,
50           -4.6796e-02]]],
51

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52      [[ 3.2582e-02,  1.0749e-02, -4.1954e-02,  2.3481e-02,  7.9805e-02,
53          -3.2564e-02],
54      [-1.0129e-02,  1.0836e-02,  2.9693e-02,  3.2025e-03, -4.2536e-02,
55          3.0045e-03],
56      [ 6.3058e-03, -3.2357e-02,  1.0005e-02,  3.0019e-02, -3.0926e-02,
57          -3.8387e-02],
58      [ 5.0602e-03,  2.1167e-02,  6.2762e-03,  2.6286e-02,  3.2144e-02,
59          1.7645e-02],
60      [-1.9601e-02,  4.1045e-02,  1.0004e-02, -2.4471e-02, -1.0165e-02,
61          -7.6924e-03],
62      [-3.8617e-02, -2.1620e-02,  1.8359e-02, -1.2443e-03, -5.9233e-02,
63          1.7056e-02],
64      [-4.4735e-03,  5.6779e-03,  8.3241e-03, -2.8883e-02,  1.2666e-02,
65          4.2973e-02],
66      [-2.5497e-02, -3.4922e-02,  4.9436e-02, -7.0054e-03, -3.6660e-02,
67          -3.4699e-02],
68      [ 2.2193e-02, -2.4997e-02, -4.0007e-02,  2.2897e-02,  2.7124e-02,
69          1.5507e-02],
70      [-9.3320e-03,  2.5782e-03,  1.4435e-02, -2.2368e-02, -1.0677e-01,
71          1.4378e-02]],
72
73      [[ 3.3989e-02,  1.7981e-03, -4.0219e-02, -1.4221e-02,  9.9310e-02,
74          1.2089e-02],
75      [-1.5836e-02, -2.1471e-02,  1.7912e-03,  3.6846e-02, -4.2144e-02,
76          -5.0264e-02],
77      [ 1.9212e-02, -1.5202e-02, -1.1652e-02,  2.0645e-02,  1.6338e-02,
78          -3.3406e-02],
79      [ 3.5625e-02,  2.3491e-02,  5.9229e-04, -1.4320e-02,  5.5157e-02,
80          2.7650e-02],
81      [-3.7838e-02,  1.4574e-02,  7.1762e-04, -3.6377e-03,  1.9236e-02,
82          2.5786e-02],
83      [-2.0315e-02, -5.2880e-03,  2.6322e-02, -8.7872e-03, -6.5245e-02,
84          -7.3839e-03],
85      [ 7.5946e-03,  2.4183e-02,  3.4703e-02, -3.9748e-02, -2.3311e-02,
86          6.4023e-02],
87      [-1.0442e-02, -2.1817e-02, -3.6961e-03,  3.1368e-02, -3.5460e-02,
88          -6.7766e-02],
89      [ 7.8273e-03,  7.3235e-03, -6.4352e-02, -1.5170e-02,  8.0027e-02,
90          -3.9284e-02],
91      [-1.6896e-02, -3.4023e-02,  4.4456e-02, -1.0631e-02, -8.4136e-02,
92          4.5353e-02]],
93
94      [[ 5.6480e-02, -6.2894e-02, -1.0786e-02,  3.5724e-02, -2.3601e-02,
95          5.3386e-02],
96      [-6.5351e-03,  2.6327e-02,  2.3450e-02,  3.7244e-02,  3.7957e-02,
97          -9.4757e-02],
98      [ 4.1822e-04, -1.3566e-01,  1.6729e-02,  6.2710e-02, -1.6384e-02,
99          -2.8333e-02],
100     [ 1.3469e-02, -2.6803e-02,  1.5821e-02, -1.1818e-03,  1.1691e-02,
101         6.5723e-02],
102     [ 1.1019e-02, -1.9738e-02,  2.7414e-02,  2.2992e-02, -2.5776e-02,
103         -1.2050e-02],
104     [ 2.4763e-03, -3.6079e-02,  2.3392e-02,  4.5837e-02,  2.6467e-02,

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105      -7.6988e-02],
106      [-3.1750e-02,  1.3655e-03,  1.2470e-02, -2.0891e-02, -1.7145e-02,
107      4.7336e-02],
108      [-1.6252e-02,  1.4886e-02, -3.3908e-03,  1.1602e-02,  2.9747e-02,
109      -5.0505e-02],
110      [ 2.2693e-02, -6.9166e-02, -1.3328e-02,  5.0020e-02, -4.2690e-02,
111      4.2718e-02],
112      [-2.7729e-02,  5.1609e-02, -1.0579e-02, -1.2472e-02,  3.3178e-02,
113      -4.9377e-02]],
114
115      [[ 1.4801e-02, -4.8090e-02, -1.3320e-02,  5.1716e-02,  4.1038e-02,
116      -4.3762e-02],
117      [-1.3575e-02,  1.5772e-02,  9.6131e-03, -3.6548e-02,  9.8821e-03,
118      2.0712e-02],
119      [ 4.7134e-03,  1.9798e-03,  6.6180e-03, -1.6347e-02, -2.2331e-02,
120      -7.9687e-03],
121      [ 3.8268e-03, -3.1876e-02, -3.6246e-02,  5.8661e-02, -3.6404e-02,
122      -1.0939e-02],
123      [ 6.3918e-05,  2.5830e-02,  2.3150e-02,  1.8586e-02, -9.4486e-03,
124      -1.3447e-02],
125      [-3.0137e-02,  9.2516e-03,  5.9200e-02, -4.2534e-02, -3.2308e-02,
126      3.2038e-02],
127      [-9.0678e-03, -3.3748e-02,  2.1461e-02,  3.4669e-02, -1.5913e-02,
128      -1.4373e-02],
129      [ 6.5261e-03,  3.7151e-02, -1.7145e-02, -3.4308e-02,  7.0612e-03,
130      4.7915e-02],
131      [ 3.1216e-02, -3.7861e-02,  8.5277e-03,  4.5494e-02,  3.8080e-02,
132      -4.1260e-02],
133      [-5.1647e-02,  4.2739e-02,  1.9673e-02, -6.6480e-02, -6.6469e-02,
134      3.6155e-02]],
135
136      [[ 6.3825e-03,  1.8330e-02, -2.5050e-02, -3.0237e-02,  5.1290e-02,
137      3.7336e-02],
138      [-1.7771e-02, -2.3943e-02, -3.0662e-02,  1.0505e-02, -4.1804e-02,
139      -2.6288e-02],
140      [-1.1268e-02, -5.5061e-03, -2.9720e-03, -2.7996e-02,  2.6197e-02,
141      6.7678e-03],
142      [ 4.1750e-04,  4.0978e-02, -3.4853e-02, -2.8867e-02,  2.1704e-02,
143      2.9822e-02],
144      [ 4.3570e-03,  2.9873e-02,  7.9196e-03, -1.9629e-02,  3.7109e-03,
145      1.2160e-02],
146      [-3.3842e-03, -6.2474e-02,  1.9763e-02,  5.1929e-02, -5.2491e-02,
147      -5.8697e-02],
148      [-1.4367e-02,  3.2151e-03,  2.5152e-02,  6.1418e-03, -9.0089e-03,
149      2.3201e-02],
150      [-8.3563e-03, -6.3644e-02, -2.2286e-04,  8.9140e-02, -1.5928e-02,
151      -1.2991e-01],
152      [ 2.9156e-02,  1.6697e-02, -4.1326e-02, -1.9598e-03,  4.4188e-02,
153      2.9923e-02],
154      [-4.4146e-02, -3.5465e-02,  4.8407e-02,  2.7974e-02, -8.5542e-02,
155      -6.5180e-03]],
156
157      [[ 3.8128e-02, -5.4455e-02, -2.5583e-02,  4.3773e-02,  1.9535e-02,

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158      -5.0362e-04],
159      [ 1.3856e-02,  7.6643e-02,  2.5912e-03, -2.3073e-02,  2.8714e-02,
160        1.6202e-02],
161      [-2.4586e-02,  1.0204e-01, -3.1662e-02, -6.7595e-02,  1.4403e-02,
162        4.3923e-02],
163      [ 1.1480e-02, -4.2273e-02,  3.2039e-03,  6.7724e-02, -7.4637e-03,
164        -5.5815e-02],
165      [-9.9504e-03,  4.3109e-02,  5.6460e-03, -3.9488e-02, -1.7515e-02,
166        5.1298e-02],
167      [-3.6934e-02,  6.1701e-02,  5.7449e-02, -3.2358e-02, -4.6462e-02,
168        3.9739e-02],
169      [-1.9788e-02, -1.1351e-02,  1.4333e-02, -1.6104e-02,  7.4954e-04,
170        -2.0050e-02],
171      [-6.2254e-03, -9.4283e-03, -1.3501e-02, -4.2459e-02, -1.1647e-02,
172        -3.4492e-03],
173      [ 2.5699e-02, -9.2446e-02, -2.3232e-02,  6.6834e-02,  2.5766e-02,
174        -5.1838e-02],
175      [-6.4873e-02,  3.3936e-02,  2.5787e-02, -5.0497e-02, -1.3870e-02,
176        7.0366e-02]],
177
178      [[ 2.4523e-02, -3.5413e-02,  2.3371e-03,  9.3518e-03, -5.8866e-02,
179        -2.0839e-03],
180      [ 2.2202e-02,  2.4601e-02,  2.8891e-02, -2.2406e-02, -8.5479e-03,
181        1.4906e-02],
182      [-2.0581e-02,  9.2914e-03, -1.5485e-03,  2.1204e-02,  8.2946e-03,
183        -5.5768e-03],
184      [-6.2484e-03,  1.2618e-02, -1.2437e-02, -1.2987e-02,  7.2316e-04,
185        -5.8049e-03],
186      [-2.1453e-02, -2.0719e-02,  1.0223e-02,  3.9638e-03,  2.8002e-03,
187        -3.1543e-02],
188      [-4.9461e-02,  3.5553e-03,  1.4239e-02, -1.6449e-02,  2.6692e-02,
189        -6.7998e-03],
190      [-3.3068e-02,  1.9393e-02,  1.5487e-02, -3.7823e-03,  1.6200e-02,
191        1.8491e-02],
192      [ 5.9951e-03,  2.4244e-03,  4.1545e-03,  2.1477e-03, -2.0528e-03,
193        2.0020e-02],
194      [-1.7913e-02, -3.2646e-02, -3.8584e-02,  4.0169e-02, -2.9375e-02,
195        -1.7939e-04],
196      [-3.2854e-02,  3.8816e-02,  1.6515e-02, -4.3136e-02,  4.1521e-02,
197        1.6670e-02]],
198
199      [[ 4.6976e-02,  3.8527e-02, -4.0731e-02, -4.1113e-02, -2.8672e-03,
200        -1.5080e-02],
201      [-1.1665e-03, -2.3842e-03,  1.6869e-02, -3.8354e-02,  2.6022e-03,
202        1.0304e-02],
203      [ 4.6005e-03,  2.8654e-02,  4.5851e-03,  6.7253e-03, -1.1570e-02,
204        -1.5880e-02],
205      [-1.9967e-04,  1.3928e-03, -6.4030e-03,  1.8964e-02, -1.8684e-03,
206        2.6456e-02],
207      [ 5.5424e-03,  7.0770e-03, -2.8977e-02,  2.6975e-04,  3.0964e-02,
208        2.4072e-02],
209      [ 2.1239e-02, -7.6888e-03,  5.6870e-02, -1.9246e-02, -3.4590e-02,
210        -9.9243e-03],

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263 =====
    =====
264 CPIKANModel                                [1, 1]                                --
265 |-----ModuleList: 1-1                    --                                --
266 |      |-----0.weight                    |-----60
267 |      |-----1.weight                    |-----600
268 |      |-----2.weight                    |-----60
269 |      |-----ChebyKANLayer: 2-1          [1, 10]                    60
270 |      |      |-----weight                |-----60
271 |      |-----ChebyKANLayer: 2-2          [1, 10]                    600
272 |      |      |-----weight                |-----600
273 |      |-----ChebyKANLayer: 2-3          [1, 1]                     60
274 |      |      |-----weight                |-----60
275 =====
    =====
276 Total params: 720
277 Trainable params: 720
278 Non-trainable params: 0
279 Total mult-adds (Units.MEGABYTES): 0.00
280 =====
    =====
281 Input size (MB): 0.00
282 Forward/backward pass size (MB): 0.00
283 Params size (MB): 0.00
284 Estimated Total Size (MB): 0.00
285 =====
    =====
286
287 进程已结束, 退出代码为 0
288

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